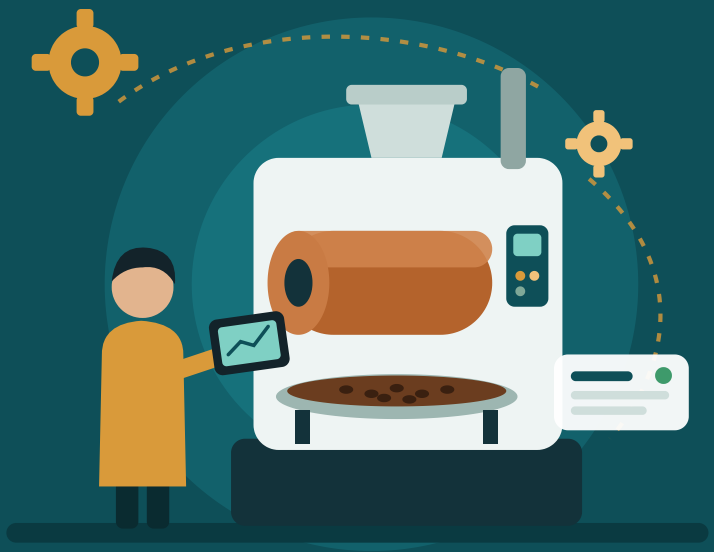
AI for machine builders

A practical guide for owners of Dutch family-owned machine builders: four questions to answer before you spend money on AI, each with five first steps you can start within 30 days.



01 Who owns the AI your company would depend on? — VENDORS & DEPENDENCE — p. 2

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What this means for you Our reading of the evidence and our advice.

From another firm A published case from a comparable company.

A vendor's view Shows a product exists, not that it works.

Still unknown What no source we checked could answer.

A team of five from HAN University of Applied Sciences

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How we made it

Our own research assistant, built on Claude, searched public sources in English and Dutch and drafted each page. We chose the sources and checked every claim against its source. Small numbers point to the sources on each page.

Educational, not consultancy: research-based starting points, not legal, commercial or safety advice. No real machine-building company is described and no information from any firm was used. Companies appear only as named in the published sources cited. AI in Business minor, HAN University of Applied Sciences, 2026.

CHAPTER 01 · VENDORS & DEPENDENCE

Who owns the AI your company would depend on?

"We buy our software from others, and more of it contains AI. Whose tools are we really relying on, and what happens if that company, its owners or its government change course?"



IN SHORT

Most AI you buy runs on American models and clouds, often inside software with foreign owners. That is no reason to avoid it, but it is a reason to know your vendors, test that you can take your data out, and sign with an exit route.

What the evidence says

- **Models and cloud.** Around 70% of foundation AI models developed since 2017 came from the US.[1] Amazon, Microsoft and Google hold about 70% of the European cloud market; all European providers together about 15%.[2]
- **A Dutch name is not Dutch ownership.** Exact has been owned by the US investment firm KKR since 2019.[3] Ridder iQ is sold by ECI Software Solutions of Fort Worth, Texas,[4] in which Leonard Green agreed to take a majority stake in 2020.[5]
- **Legal reach.** Under the US CLOUD Act, US authorities can demand data from US companies wherever it is stored; Microsoft France said it could not guarantee French customer data would never be handed over.[6]
- **Switched off.** In June 2026 a US export-control directive made Anthropic suspend access to two AI models for all non-US persons;[7] it was lifted about three weeks later.[8]
- **Features and inputs.** Meta held back coming multimodal models from the EU in 2024;[9] in 2025 the Dutch AP warned against entering personal data into DeepSeek, which stores data in China.[10]
- **EU law helps.** Under the Data Act you may switch cloud providers, with no switching fees from 12 January 2027; a medium-sized machine builder must tell users what data its connected products generate.[11] High-risk AI rules now apply from 2 December 2027 and 2 August 2028.[12]

What this means for you

Firms your size buy AI inside software. Ask every vendor for its list of subprocessors, keep a manual way to plan the workshop, and never let one tool be the only way to run it.

Example: a dependency check before a planning-tool trial

1 Goal
A go/no-go with an exit route, within two weeks.

2 Data
List what the tool gets, and what it must never get.

3 Export
Check you can export the data again first.

4 Compare
Score two vendors 0–2 on five dependence points.

5 Weigh
Less dependence can beat a few extra features.

6 Start
Exit clause, four-week trial next to current planning.

~70%
of foundation AI models since 2017 came from the US[1]

~70%
of Europe's cloud market is held by Amazon, Microsoft and Google[2]

45%
of Dutch firms with 50–250 staff used AI in 2025[13]

2027
no more cloud switching fees from 12 January[11]

Your first five steps

1. Make a vendor list: every tool that touches design, planning, customer data or machine software.
   Owner + office manager · this week
2. Ask each vendor in writing: where is our data, who else touches it, is it used for training?
   Office manager
3. Run one export test on your most important system.
   Planner REVERSIBLE
4. Score AI-tool candidates on dependence as well as function and price.
   Owner
5. Put an exit clause in your next contract: export, notice period, no switching fees.
   Owner · next renewal

Responsible AI

Data: where data sits is not who controls it; sign a data-processing agreement for personal data.

A person decides: the tool advises, a named planner decides.

Customers: tell them what your machines collect; the Data Act lets them ask.

SOURCES

1. **Draghi (2024).** The future of European competitiveness, Part A, p. 24. EU-commissioned policy report.
2. **Synergy Research Group (Jul 2025).** European cloud market share. Specialist estimates.
3. **Apax Partners (Feb 2019).** KKR to acquire Exact. Seller's announcement.
4. **ECI Software Solutions.** Ridder iQ (NL). Vendor's own page.
5. **Leonard Green & Partners (Nov 2020).** ECI to be acquired. Buyer's announcement.
6. **The Register (Jul 2025).** Microsoft "cannot guarantee" data sovereignty. Senate testimony.
7. **Anthropic (Jun 2026).** Statement on model access. Affected vendor.
8. **Al Jazeera (Jul 2026).** US lifts restrictions on Anthropic's models. Independent news.
9. **Axios (Jul 2024).** Meta won't offer future multimodal AI in EU.
10. **NOS (Jan 2025).** AP waarschuwt voor DeepSeek. Dutch public broadcaster; in Dutch.
11. **European Commission (Dec 2025).** Data Act explained. Explainer, not the legal text.
12. **White & Case (Jul 2026).** EU AI Omnibus enters into force. Law-firm summary.
13. **CBS (Dec 2025).** Bedrijven gebruiken AI vaakst voor marketing of verkoop.

Drafted with our research assistant from claims the team accepted and checked against their sources.

CHAPTER 02 · TRUST & AI SAFETY

Can we trust AI with our drawings, data, machines and people?

"Can we trust an AI tool with our drawings, our customers' data and our planning, without creating a data, safety or people problem?"



IN SHORT

Do not ask whether AI is safe in general. For each use, decide what data may go in, whether AI may only advise or also act, and how staff tell you they use it. Start small and reversible, with a named person responsible.

What data may leave the building?

Green · public
Brochures, published manuals and specifications.
Approved AI tools only.

Amber · controlled
Lead times without names, anonymised service tickets.
Owner approval, minimal data, vendor agreement.

Red · never outside
Customer drawings, recipes, quotes and prices, staff data, machine code, passwords.
Never in public tools.

When may AI act?

Green · assists
Summaries, translations, draft emails.
A person checks.

Amber · recommends
Workshop sequence, late-order flags.
A named planner approves.

Red · never alone
Machine and safety settings, delivery promises, staff decisions.
A person decides.

What the evidence says

- **Leaks.** Pasting personal data into a chatbot can itself be a data breach; reporting it to the AP and those affected is often mandatory.[2] In 2025 the AP received dozens of such reports, often involving free chatbot versions.[3]
- **Staff.** Since 2 February 2025 the AI Act bans AI that infers employees' emotions; AI that allocates tasks or monitors staff is high-risk,[5] with obligations from 2 December 2027.[6]
- **Machines.** From 20 January 2027 the Machinery Regulation treats self-learning safety functions as high-risk, needing third-party assessment.[7,8]

A vendor's view

OpenAI states it does not train on data from its business and API products by default; consumer versions have separate terms.[4] A Microsoft–LinkedIn survey found 75% of knowledge workers use AI at work, 78% with their own tools.[9]

What this means for you

Keep AI out of machine control and safety functions until you know which rules apply. Treat quiet AI use as a training issue: publish a one-page rule with a no-blame way to report an accidental upload.

49%
of Dutch firms that held back from AI named privacy[1]

42%
named unclear legal consequences, such as liability[1]

78%
of AI-using office workers bring their own tools (vendor survey)[9]

2027
new Machinery Regulation applies from 20 January[7]

Your first five steps

1. Make a one-page data map: drawings, customer, planning, machine and staff data as green, amber or red.
   Owner + office manager REVERSIBLE
2. Publish a one-page AI rule for staff: approved tools, forbidden data, who to ask, how to report an upload.
   Owner
3. Choose one amber use where AI only advises; name its owner, success measure, stop rule and end date.
   Owner + planner
4. Write two approval moments, such as a planner approving every late-order flag.
   Operations manager
5. Ask your machine-software supplier: does our machine control use AI or self-learning functions?
   Owner

Responsible AI

Data: classify first, use as little as possible, get a written data-processing agreement.

Reliability: a tool that cannot explain a flag may not be the only basis for a decision.

People: AI supports the work; it is not there to monitor staff.

SOURCES

1. **CBS (Dec 2025).** Bedrijven gebruiken AI vaakst voor marketing of verkoop. Statistics Netherlands; self-reported.
2. **Autoriteit Persoonsgegevens.** Gebruik AI-chatbot kan leiden tot datalekken. Dutch regulator; guidance.
3. **ICT Magazine (Dec 2025).** AP: meer datalekken door gratis chatbots. Trade press reporting the AP.
4. **OpenAI (Jan 2026).** Enterprise privacy. Vendor's own commitment.
5. **Regulation (EU) 2024/1689 (AI Act).** Art. 4, Art. 5(1)(f), Annex III(4). Primary EU law; amended 2026.
6. **White & Case (Jul 2026).** EU AI Omnibus enters into force. Law-firm summary.
7. **Baker McKenzie.** Machinery Regulation. Law firm; legal text is (EU) 2023/1230.
8. **EU-OSHA.** Regulation 2023/1230/EU: machinery. EU agency summary.
9. **Microsoft & LinkedIn (May 2024).** Work Trend Index. 31,000 people, 31 countries; vendor with an interest.

Drafted with our research assistant from claims the team accepted and checked against their sources.

CHAPTER 03 · KNOWLEDGE GAPS

What do we not know yet, and what should we find out first?

"What do we not yet know about AI, and which gap should we close first before we spend money on it?"



IN SHORT

Owners ask for examples, but the evidence points to an earlier gap: most firms do not yet know their own data. Before any AI trial, find out how reliable your own delivery dates are.

What the evidence says

- **Firms your size are moving.** AI use among Dutch firms with 50–250 staff rose from 20% in 2023 to 45% in 2025, but only 16% of Dutch industry firms used AI in 2025.[1] Across the EU, 30.36% of medium-sized firms used AI.[2]
- **The barriers firms name.** Lack of experience (73%), privacy (49%) and unclear legal consequences (42%).[1] The EU picture is the same: expertise 70.89%, legal 52.52%, privacy 48.83%.[2] Time, maintenance costs and skills gaps also hold SMEs back.[5]
- **The rules.** Half of Dutch entrepreneurs do not know the AI Act and only 7% are well informed; 44% of firms with 10+ staff expect it to affect them.[3]
- **Inside the factory.** In 80% of SME manufacturers surveyed, data and sensor use is not developed enough, and 83% do not know which processes suit AI. Owners say they want examples, step-by-step plans and investment help.[4]

From another firm

In two German build-to-order manufacturers, sales estimated delivery times from experience and added a large buffer, so the date first promised was at high risk of not matching capacity.[6]

What this means for you

The gap that matters most is your own delivery-date performance. Without it you cannot choose an AI trial, and you cannot tell whether one worked. No AI is needed to measure it: an ERP export and a spreadsheet will do.

Still unknown

How reliable your own dates are, and whether your ERP records promised and actual dates for every order.

Example: a delivery-date baseline

1 Goal
On-time share and the top three reasons for a miss.

2 Data
Order, promised and actual dates; changes; supplier delays.

3 Steps 3 to 6
Prepare, build, test, start: see chapter 4.

45%
of Dutch firms with 50–250 staff used AI in 2025, up from 20%[1]

16%
of Dutch industry firms used AI in 2025[1]

73%
of firms that held back name lack of experience[1]

83%
of SME manufacturers do not know which processes suit AI[4]

Your first five steps

1. Name one person who owns "how reliable are our delivery dates?"
   Owner · week 1
2. Export 12 months of orders with promised and actual dates; note missing fields.
   Planner or ERP user · weeks 1–2 REVERSIBLE
3. Count on-time deliveries; list the top three reasons for a miss.
   Owner with planner · week 3
4. Write a one-page rule: no customer names, drawings or prices in public chatbots.
   Office manager · week 2
5. Check the AI Act's user duties against the AI tools your staff already use.
   Owner with office manager · week 4

Responsible AI

Data: the baseline uses your own order data; keep it inside the firm.

Rules: AI Act users must check whether a tool falls under the law, agree terms with suppliers, be open with customers and train staff.[3]

A person decides: people still set every delivery date.

SOURCES

1. **CBS (Dec 2025).** Bedrijven gebruiken AI vaakst voor marketing of verkoop. Statistics Netherlands; self-reported, all sectors.
2. **Eurostat (Dec 2025).** Use of artificial intelligence in enterprises. EU statistical office; EU totals.
3. **KVK (Sep 2025).** Ondernemers zetten nauwelijks stappen rond Europese AI-wetgeving. Chamber of Commerce; 651 respondents.
4. **FME et al. (Apr 2026).** Mkb-maakindustrie zet digitaliseringsstappen. Sector bodies with Fontys; Brabant focus.
5. **OECD (Apr 2026).** Empowering SMEs in the age of AI. 2,000+ SMEs; abstract read; non-representative.
6. **Rokoss et al. (2024).** Delivery time determination using machine learning in small batch production. Peer-reviewed; two German firms.

Drafted with our research assistant from 20 claims our tester accepted on 10 Oct 2026; the tester opened three sources, our Dutch reader checked the translations, and every quote was matched with its source. Desk research only.

CHAPTER 04 · USE CASES

What do firms like ours use AI for, and what is worth trying first?

"What do firms like mine actually use AI for, what pays off, and which one use should we try first?"



IN SHORT

Most firms use AI in the office, and we found no measured case of AI planning at a Dutch machine builder. The best-documented option for your problem: a realistic delivery date at the quote, tested on paper first.

What the evidence says

- **Mostly the office.** Of Dutch firms using AI, 35% use it for marketing or sales, 32% for administration including planning, 18% for production or service, and 4% for logistics.[1] EU manufacturers mostly use it for marketing and sales (30.41%) and administration (27.05%).[2]
- **Mostly off-the-shelf.** SMEs keep taking up ready-made AI tools, but deeper, secure integration into operations remains uneven.[5] Dutch SME manufacturers increasingly choose small, feasible steps, and involving staff from the start is key.[4]
- **What is advised.** The Asian Productivity Organization recommends predictive maintenance, visual inspection, digital twins, human-in-the-loop work and federated learning.[8]

A vendor's view

PayPal says AI agents can automate reordering, invoicing, financial records and compliance reports.[6]

From another firm

Two German make-to-order firms predicted delivery dates with machine learning. A date estimated at the quote lowered the risk of late delivery but had 35% more error than the planners' later date.[7]

What this means for you: our pick

Try a realistic delivery date at the quote, starting with a rule of thumb from your own history. A machine builder probably has too few orders for machine learning at first.

Example: a realistic date at the quote

1 Goal
Promise dates you can keep; agree a stop rule.

2 Data
Desired, promised and actual dates; route; outside work.

3 Prepare
Link each order to changes and supplier delays.

4 Build simple
A rule of thumb before any machine learning.

5 Test
Three months: estimate vs planner vs real date.

6 Start
The estimate advises; a person sets the date.

35%
of Dutch AI users use it for marketing or sales[1]

4%
use it for logistics, the least common purpose[1]

+35%
more error in a quote-stage date than the planners' later date[7]

16,361
orders in one German firm's data behind that result[7]

Your first five steps

1. Choose this use case and name the planner as its owner.
   Owner · week 1
2. Start from the chapter 3 baseline: on-time share and top reasons for misses.
   Planner · weeks 1–3
3. For the next 10 quotes, write the planner's date beside a simple estimate; compare both with the real date later.
   Planner with sales · from week 2 REVERSIBLE
4. Ask two planning-software vendors for references from machine builders, with before-and-after figures.
   Owner · week 3
5. Write down what data a vendor may access, and that a person always sets the date.
   Office manager · week 4

Responsible AI

A person decides: AI advises; the planner commits the date.

Vendors: AI Act users must agree in writing what a supplier may do with their data.[3]

Still unknown: whether a date-setting planning tool falls into a higher risk class.

SOURCES

1. **CBS (Dec 2025).** Bedrijven gebruiken AI vaakst voor marketing of verkoop. Statistics Netherlands; self-reported.
2. **Eurostat (Dec 2025).** Use of artificial intelligence in enterprises. EU statistical office; EU totals.
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6. **OECD Cogito blog (Sep 2025).** Agentic AI for Small Business Growth. Written by PayPal; vendor's view.
7. **Rokoss et al. (2024).** Delivery time determination using machine learning in small batch production. Peer-reviewed; two German firms.
8. **Asian Productivity Organization (2026).** Leveraging AI for Smart Manufacturing in SMEs. Guidance, not results.

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